Computer Networks

Homework 2 (CHAP. 2)

1. List five nonproprietary Internet applications and the application –layer protocols that they use.

2. Caching, for example DNS RR caching or Web page caching, is one of useful techniques in computer systems. Describe what are possible benefits to use caching mechanism. And, at the same time, what are possible side-effects (in the bad sides) we much be aware of when caching mechanisms are applied?

3. DNS. (Application layer)

   In general, there are two mechanisms for DNS domain name resolving (that is, find out the IP of a given domain name), iterative and recursive. Refer to Figure 1, suppose now the host "cis.poly.edu" would like to know the IP addresses for the domain name "gaia.cs.umass.edu".

   (a). Please draw DNS query message flow (both for dns request and reply) for the two mechanisms individually. (Assume no cache results in any of the hosts or servers.)
   (b). How DNS servers know the IP addresses for root DNS servers?
   (c). Why not centralize DNS system?
   (d). By default, DNS servers use iterative mechanism for name resolving. What are possible reasons?
   (e). For each request (both in iterative/recursive), we might need to query root DNS server. The number of requests should be huge. Why we, in practices, just need few number of root servers (e.g. 14-16 worldwide) to handle the requests.
   (f). How DNS cache works? Why use DNS cache?

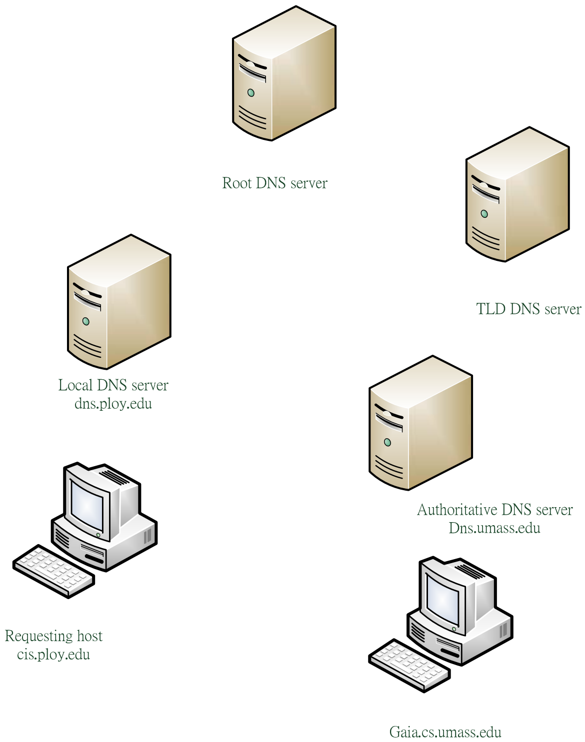

Fig. 1 DNS query: interaction
of the various DNS servers

Figure 1: DNS request scenario

4. DNS VULNERABILITIES.

   We have seen that DNS is a critical component of the Internet infrastructure, and so it is quite nature to be an attacking target. How can DNS be attacked? For example, you can find different types of DNS Attacks from internet (e.g. https://www.paloaltonetworks.com/cyberpedia/what-is-a-dns-attack.) Please give 3 possible DNS attach approaches. For each of them, please give a short description about how it works.

5. Streaming multimedia: DASH (Dynamic, Adaptive Streaming over HTTP)
   (a). What's main reason to send streaming data over HTTP ?
   (b). Why DASH chops the video file into small pieces (chunks)?
   (c). How DASH deal with heterogeneous clients (that is, bandwidth available for different users might not be the same, some are with rich bandwidth some are not) ?
   (d). How DASH adapts to highly dynamics of internet traffics (for example, end-to-end bandwidth and packet delays are highly unpredictable)?

6. Content Distribution Networks (CDNs)

(a) what are the main services provided by CDN

(b) what are possible benefits to use CDN for a content provider, eg. NetFlix

(c) CDN can effectively mitigate DDoS attacks, why?

7.OTT media service vs. IPTV

(a) currently, media services can be provided via OTT or IPTV, what are the difference between them (you may want to refer to https://otter-video.com/blog/iptv-vs-ott/)

(b) give an example of OTT media service provider in Taiwan (or your country), and an example for IPTV provider.

(c) In your opinion, which one (OTT or IPTV) might win in media service market? why?

8. Internet tools: nslookup, traceroute, whois

(a) Please use **nslookup** to check what is the IP address (or addresses) of www.facebook.com and www.google.com. Write down what you found and compare your answer with your classmates and check if there are any differences.

(b) Please use **traceroute (or tracert)** command to check the routing path to the server

(c) Please use **whois** command to check the owner of the IP address.